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On the use of transfer learning in nature-inspired algorithms: a systematic review

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Abstract
Transfer Learning (TL) has gained significant traction in machine learning, especially in deep learning contexts. However, its integration with Nature-Inspired Algorithms (NIAs) remains fragmented, with limited understanding of strategies, challenges, and outcomes. This paper presents the first systematic review focused exclusively on the use of TL in NIAs, excluding deep learning approaches. Major challenges include dealing with domain/task similarity, avoiding negative transfer, selecting what and when to transfer, and adapting TL mechanisms to population-based search paradigms. To address these issues, we conducted a structured analysis of 47 primary studies, categorizing them by TL strategies, learning paradigms, and algorithmic goals. Our findings reveal recurring patterns, highlight open research gaps, and propose future directions for developing robust TL-based NIAs. This review provides a foundation for researchers interested in designing adaptive, efficient, and knowledge-guided metaheuristics for complex optimization tasks.

Keywords Transfer learning · Nature-inspired algorithms · Evolutionary algorithms · Swarm intelligence · Optimization · Systematic review · Knowledge transfer · Bioinspired computing · Machine learning

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1 Introduction

The increasing complexity of optimization problems in real-world applications has driven the development of adaptive and intelligent computational techniques. Among these, nature-inspired algorithms (NIAs) have gained prominence due to their flexibility, scalability, and ability to operate in dynamic, high-dimensional environments. At the same time, transfer learning (TL) has emerged as a promising approach to enhance learning systems by reusing knowledge across tasks and domains. Combining these two paradigms opens new opportunities to solve challenging problems more efficiently and with stronger generalization capabilities.

1.1 Motivation and problem statement

Natural Computing (NC) studies and models natural processes and phenomena to develop nature-inspired algorithms that solve problems inadequately addressed by traditional methods. Nature-inspired algorithms, such as Evolutionary Algorithms (EA) and Swarm Intelligence (SI), illustrate how understanding the behavior of living organisms–whether collectively or individually–can inspire technological and scientific advances De Castro (2006). NC algorithms have been successfully applied not only to computational tasks such as image and document classification but also to industrial systems Ardeh et al. (2021), logistics problems Zheng et al. (2022), and manufacturing processes Zhou et al. (2023).

NC systems are capable of adapting to their operating environments. For example, Genetic Algorithms (GA) evolve populations through genetic variation, while Particle Swarm Optimization (PSO) algorithms allow individuals to adjust their behavior based on social interactions Siddique and Adeli (2015). This adaptability is a key feature that makes nature-inspired algorithms suitable for complex, dynamic problems.

Transfer Learning (TL), in turn, enables knowledge from a model trained on one task or domain (source) to be reused to accelerate learning or improve performance on another task or domain (target) Pan and Yang (2009). Although widely used in deep learning, TL holds promising potential in other paradigms, including nature-inspired algorithms. This relevance is particularly clear in optimization contexts, where understanding why knowledge transfer improves convergence and generalization is just as important as designing the transfer mechanism itself.

By integrating TL into NIAs, it is possible to improve generalization, reduce learning time, and decrease computational costs. For example, an Ant Colony Optimization (ACO) algorithm could benefit from reusing a previously learned pheromone map instead of reconstructing it from scratch at every iteration Singh and Pillay (2022).

Although transfer learning is well established in deep learning contexts, its application in NIAs remains underexplored. Existing reviews mostly focus on neural networks, overlooking adaptations required for TL in evolutionary and swarm-based algorithms. To the best of our knowledge, no prior systematic review has specifically investigated the intersection of TL and NIAs outside of deep learning. This work addresses that gap by analyzing the main strategies, objectives, and challenges involved in this integration.

The motivation behind this review stems from the increasing complexity of real-world optimization problems and the need for more efficient, generalizable, and adaptive metaheuristics. While nature-inspired algorithms are robust, they often suffer from high compu-

tational costs, limited re-usability, and sensitivity to initial configurations. Transfer learning presents an opportunity to overcome these limitations by transferring prior knowledge to new problem settings. Nevertheless, open challenges such as domain/task mismatch, the risk of negative transfer, and the precise definition of transferable components remain crucial obstacles to be addressed.

The objective of this paper is to conduct a systematic review focused exclusively on the use of Transfer Learning in Nature-Inspired Algorithms (excluding deep learning), intending to identify key strategies, challenges, and research trends that may support the development of more adaptive and reusable metaheuristics in complex optimization scenarios.

Furthermore, this review not only describes existing approaches but also discusses why certain transfer learning strategies tend to be more effective than others when integrated with NIAs. These differences often stem from algorithmic properties, such as adaptability, representation of knowledge, and robustness to dynamic environments, which directly influence transferability and performance gains.

1.2 Contributions

This paper makes the following main contributions:

- It presents the first systematic review that exclusively investigates the integration of transfer learning in nature-inspired algorithms, excluding deep learning approaches, thus addressing a specific gap in the literature.
- It proposes a novel classification framework for TL strategies used in NIAs, organizing them based on learning paradigms (inductive, transductive, unsupervised), transfer components (what, when, how), and algorithmic goals.
- It identifies and synthesizes key technical challenges in combining TL with NIAs, such as domain/task mismatch, risk of negative transfer, and difficulties in defining transferable components.
- It reveals recurring patterns across 47 primary studies and provides a comparative analysis of how different TL strategies are implemented in evolutionary and swarm-based algorithms.
- It outlines open research directions and future opportunities for designing adaptive and generalizable TL-based metaheuristics in complex optimization scenarios.

It is also important to acknowledge that most surveyed studies are concentrated in specific application domains. Nevertheless, the identified challenges–particularly domain/task similarity and the risk of negative transfer–are general issues that extend to a wide range of optimization problems. This reinforces the broader relevance of the review and highlights directions for future research beyond the contexts analyzed.

1.3 Organization

This review is divided into six sections, including the introduction:

- Section 2 presents the theoretical background on transfer learning and nature-inspired algorithms.

- Section 3 describes the review protocol.
- Section 4 presents the results of the systematic review.
- Section 5 discusses research trends, challenges, and future directions.
- Finally, Section 6 concludes the paper.

2 Theoretical background

This section provides an overview of transfer learning, covering its main concepts, strategies, and research challenges, as well as fundamental principles of natural computing. It also introduces the algorithms found in the primary studies, along with the main problems addressed by these algorithms.

2.1 Transfer learning: core concepts

Transfer learning operates on the principle that knowledge acquired in one context can be reused in another, facilitating learning in new but related scenarios. Just as humans rely on prior knowledge to acquire new skills, computational models can transfer knowledge from previously solved tasks to improve performance in novel environments Pan and Yang (2009).

TL is characterized by the adaptation of a model or system from a source domain, denoted as $\mathcal{D}_S$, with a corresponding source task $\mathcal{T}_S$, to improve the performance of a target task $\mathcal{T}_T$ in a target domain $\mathcal{D}_T$, where $\mathcal{D}_S \neq \mathcal{D}_T$ and $\mathcal{T}_S \neq \mathcal{T}_T$. Formally, transfer learning can be represented as follows:

$$\mathcal{D}_S, \mathcal{T}_S \rightarrow \mathcal{D}_T, \mathcal{T}_T$$

In this context, $\rightarrow$ corresponds to the process of knowledge transfer, where the primary objective is to facilitate the learning of $\mathcal{T}_T$ by capitalizing on the insights and experience obtained from $\mathcal{T}_S$.

2.1.1 Transfer learning: strategies and challenges

Transfer learning can be classified according to various criteria Zhuang et al. (2020). One common classification, based on Pan and Yang (2009), divides TL techniques into three main categories of learning strategies:

1. **Inductive-learning:** Source and target tasks differ, while domains may be the same or different. The goal is to generalize learned knowledge to a new task.
2. **Transductive-learning:** Transductive Transfer Learning: Tasks remain the same, but domains differ. The goal is to adapt the model to a new data distribution.
3. **Unsupervised-learning:** Both tasks and domains differ, with no labeled data available in the target. Useful in scenarios with limited annotation.

In addition to learning strategies, Pan and Yang (2009) describes four main *transfer strategies*, which focus on the techniques used to transfer knowledge from the source domain to the target domain:

1. **Instance-transfer:** Reweights instances from the source to be reused in the target domain.
2. **Feature-representation-transfer:** Learns a shared representation that minimizes distribution differences between domains.
3. **Parameter-transfer:** Transfers model parameters or structures.
4. **Relational-knowledge-transfer:** Transfers relational patterns or structures across domains.

TL is not only about learning or transfer strategies, but also about understanding why some approaches are more effective than others in practice. In this context, three key research issues must be addressed Pan and Yang (2009):

1. **What to transfer:** This issue involves identifying the parts of knowledge that can be transferred from the source task to the target task (e.g., weights, features, models).
2. **How to transfer:** This involves determining the method or algorithm used to apply knowledge from the source task to the target task (e.g., at initialization, during training, after convergence).
3. **When to transfer:** Deciding when it is appropriate to apply transfer learning, considering factors such as domain similarity and dataset characteristics (e.g., fine-tuning, reweighting, domain adaptation).

2.1.2 Developmental process

There are a series of procedures that must be taken into account for the application of Transfer Learning. Its development process typically encompasses the following main steps:

1. **Source domain and task identification:** Begin by identifying a suitable source domain $\mathcal{D}_s$ and the source task $\mathcal{T}_s$ that possess relevant knowledge applicable to the target domain and task.
2. **Feature extraction and representation learning:** Extract and represent informative features from data from the source domain, ensuring that these features are relevant and transferable to the source and target tasks.
3. **Model adaptation:** Adapt the model, often pre-trained on the source task, to the target task, while considering any discrepancies in data distributions and label spaces between the source and target domains.
4. **Fine-tuning and training:** Fine-tune the adapted model on the data of the target domain to optimize its performance for the target task. This step often includes adjusting model hyper-parameters and learning rates.
5. **Evaluation and monitoring:** Continuously evaluate the performance of the transfer learning model on the target task and make the necessary adjustments based on performance metrics such as accuracy, precision, and recall.

The effectiveness of transfer learning is heavily dependent on the careful execution of these steps, tailored to the specific problem and the domain under consideration. However, unresolved issues such as accurately measuring domain/task similarity and avoiding negative transfer remain fundamental challenges across different optimization contexts.

2.2 Nature-inspired algorithms: overview

Natural computing is a field of computer science that draws inspiration from natural systems to develop novel computational methods for solving complex problems. This interdisciplinary area integrates computer science with other scientific domains such as chemistry, physics, and biology Siddique and Adeli (2015). According to De Castro (2007), natural computing can be divided into three main categories, as summarized in Fig. 1:

1. **Nature-inspired algorithms (NIA)**: Methods that use inspiration from nature to develop new techniques for problem solving.
2. **Simulation of natural phenomena in computing**: The use of computational systems to synthesize natural phenomena.
3. **Computing with new nature materials**: Methods that use natural materials for computation.

Recently, there has been a notable research focus on Artificial Neural Networks (ANNs), a method within the NIA category, driven by their success in emerging machine learning approaches such as deep learning Barthi et al. (2020) and natural language processing (NLP) Young et al. (2018). Thus, research emphasizing other natural computing categories may identify gaps and foster new paradigms within the field.

This systematic review sheds light on nature-inspired algorithms that stand to benefit from transfer learning. Most algorithms examined in the base studies belong to the categories of evolutionary algorithms and swarm intelligence. The following subsections present key concepts from these two areas.

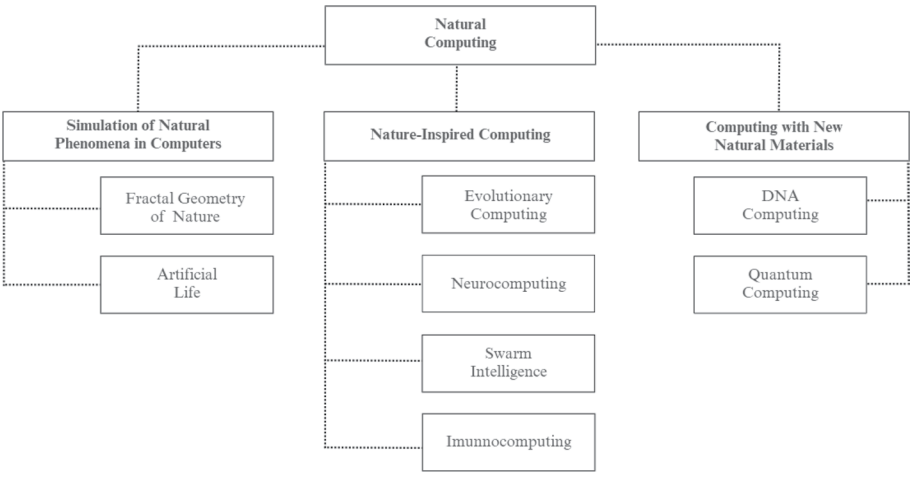


Fig. 1 Main natural computing categories and approaches

2.2.1 Evolutionary algorithms

Most algorithms employing transfer learning in this review fall under Evolutionary Computing. Inspired by biological evolution, evolutionary algorithms (EAs) simulate processes such as reproduction, mutation, selection, and inheritance to evolve a population of candidate solutions over time De Castro (2007). In Fig. 2, it was summarized the NIAs detected in base studies.

The reviewed studies include various EA variants, such as Genetic Algorithms (GA), Genetic Programming (GP), Multiobjective Evolutionary Algorithms (MOEA), Multifactorial Evolutionary Algorithms (MFEA), Evolution Strategies (ES), and Differential Evolution (DE) Siddique and Adeli (2015).

GA is a general-purpose search algorithm based on natural selection, typically using binary encoding, fixed-size populations, crossover, mutation, and probabilistic selection De Castro (2007). GP evolves symbolic programs represented as trees of terminals and functions to solve specific tasks (Mayr 1988; De Lacerda and De Carvalho 1999; Baeck et al. 2018).

MOEAs optimize multiple, often conflicting objectives simultaneously, using dominance-based criteria to evaluate solution quality Fliege et al. (2009). Variants include:

- DMOEA, for problems with time-varying objectives (DMOPs), where solutions must adapt to changing conditions (Ye et al. 2022a; Oyama et al. 2010).
- MMOEA, for locating multiple Pareto-optimal solutions in multimodal landscapes Ji

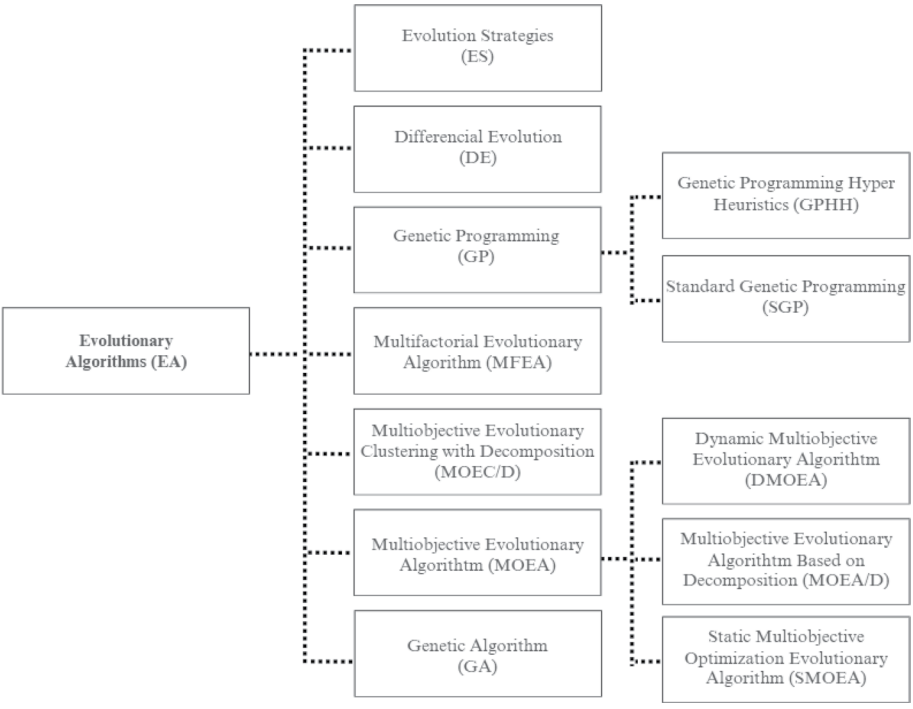


Fig. 2 Evolutionary algorithms in the base studies

et al. (2022).
- MOEA/D, which decomposes the problem into simpler subproblems Fan et al. (2020).
- SMOEA, which performs optimization without updating the population during evolution Yao and Wang (2023).

MFEA, designed for multitask optimization (MTOP), assigns individuals to tasks and evaluates them within a unified search space, promoting knowledge transfer through crossover and mutation Ma et al. (2020). MOEC/D, derived from Multiobjective Evolutionary Clustering, improves convergence by evolving clusters of promising solutions guided by community modularity Liu et al. (2019).

ES adopts a mutation-based approach where offspring replace parents if they outperform them; otherwise, parents are retained. These algorithms often leverage knowledge from previous solutions Lai et al. (2023). DE perturbs individuals using scaled differences between randomly selected solutions instead of probabilistic crossover, enhancing exploration and aiding in escaping local optima Siddique and Adeli (2013).

2.2.2 Swarm intelligence

Another nature-inspired computing approach observed in the base studies was that of Swarm Intelligence, as summarized in Fig. 3. SI arises from the collective behavior of decentralized, self-organized agents interacting locally with each other and their environment. Classic examples include flocks of birds, schools of fish, or swarms of bees, where group behavior leads to complex coordinated actions absent in individuals Eberhart et al. (2001).

SI research is commonly divided into two lines: 1) Algorithms inspired by social insects, and 2) Models based on human collective behavior. In both, coordination arises from interactions among simple agents without centralized control De Castro (2007).

Particle Swarm Optimization (PSO), a well-known SI algorithm, models particle movements through a search space based on their own best experiences and those of neighbors, promoting social learning and collective optimization De Castro (2007). Ant Colony Optimization (ACO) is based on how ants find shortest paths by depositing pheromones that reinforce favorable routes, enabling the swarm to converge to optimal solutions De Castro (2007). The Artificial Bee Colony (ABC) algorithm is inspired by honeybee foraging behavior, involving employed bees, onlookers, and scouts that explore, evaluate, and share information about food sources to guide optimization Zhou et al. (2023).

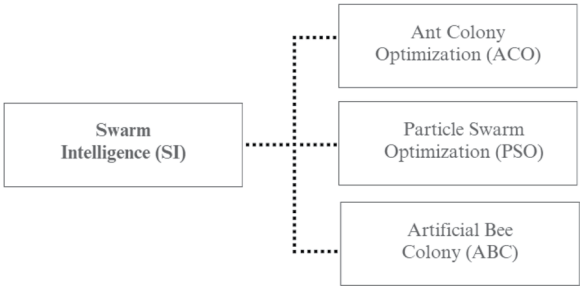


Fig. 3 Swarm intelligence algorithms in the base studies

2.3 Common problems solved by nature-inspired algorithms

Nature-inspired algorithms address a wide range of complex optimization problems, many of which are NP-hard or involve conflicting objectives, high-dimensional search spaces, or dynamic conditions. The main categories identified in the reviewed studies include:

- Multiobjective Optimization Problems (MOPs): Optimize two or more conflicting objectives simultaneously, evaluated using Pareto optimality Fliege et al. (2009).
- Dynamic Multiobjective Problems (DMOPs): Extend MOPs to dynamic environments requiring adaptive strategies (Jiang et al. 2017; Farina et al. 2004).
- Multi-Scenario MOPs (MSMOPs): Address trade-offs across different predefined scenarios within a shared decision space Zhao et al. (2023).
- Multimodal Multiobjective Problems (MMOPs): Locate multiple equivalent Pareto-optimal solutions to improve solution diversity Yue et al. (2019).
- Expensive Optimization Problems (EOPs) and Expensive Dynamic MOPs (EDMOPs): Involve costly evaluations or simulations under static or dynamic conditions Li et al. (2022).
- Multifactorial Optimization Problems (MFOPs) and Multitasking Optimization Problems (MTOPs): Solve multiple tasks simultaneously or cooperatively, exploiting task synergies (Ma et al. 2020; Lu et al. 2023).
- Cloud Resource Allocation Problems: Include scenarios such as Multiobjective Location-Aware Service Brokering (MOLSB) and Manufacturing Cloud Service Allocation (MCSA) focusing on cost, latency, and scalability (Lu et al. 2023; Zhou et al. 2023).
- Bi-Level Optimization Problems (BLOPs): Hierarchical problems where lower-level solutions constrain upper-level ones, often computationally demanding Chen et al. (2019).
- Function Approximation: Model fitting tasks common in regression and surrogate modeling Alghamdi and Heckendorn (2021).
- Classification and Regression: Supervised learning tasks involving label assignment or value prediction (Nyathi and Pillay 2021; Al-Helali et al. 2020).

Additionally, discrete optimization problems addressed include:

- 1D Bin Packing Problem (1BPP): Packing items of different sizes into containers with fixed capacity, minimizing the number of bins Singh and Pillay (2021).
- Traveling Salesman Problem (TSP): Finding the shortest route through a set of cities without repetition Zheng et al. (2022).
- Movie Scene Scheduling Problem (MSSP): Sequencing scenes to minimize production costs based on multiple variables Singh and Pillay (2022).
- Quadratic Assignment Problem (QAP): Assigning facilities to locations with cost minimization based on pairwise interactions Lu et al. (2023).
- Uncertain Capacitated Arc Routing Problem (UCARP): Finding routes under uncertain arc demands and capacities Ardeh et al. (2019).

These diverse problem classes highlight the versatility of nature-inspired algorithms and reinforce the relevance of applying transfer learning techniques to enhance their performance across dynamic, multiobjective, and high-complexity optimization scenarios.

3 Survey protocol

This paper presents the first systematic survey on the state of the art regarding the use of transfer learning in nature-inspired algorithms. The aim is to understand, identify, and analyze the benefits, challenges, and gaps in their combined application. The study follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines Page et al. (2021), which encompass procedures including formulation of research questions and search terms (search strings), selection of research sources and management software, and definition of inclusion and exclusion criteria. By explicitly adhering to these standards, the review ensures transparency, replicability, and methodological rigor. These processes are detailed in the following sections.

3.1 Research questions and terms

This systematic review addresses the following questions:

- **Question 1:** Which natural computing algorithm uses transfer learning, and what are the main issues ("What to transfer?", "How to transfer?", and "When to transfer?") in the application of TL to natural computing algorithms?
- **Question 2:** What problem of the natural computing algorithm is solved by transfer learning?
- **Question 3:** What problem is being solved by the nature-inspired algorithm, how is performance evaluated, and which algorithms are used for performance comparison?

To retrieve articles relevant to these questions, keywords and their synonyms were selected. Search queries were constructed using logical operators such as OR, AND, and NOT (or their equivalents) for compatibility with search engines. The following search strings were used:

- **Query 1:** "transfer learning" + "nature-inspired algorithms" -"deep learning"
- **Query 2:** "transfer learning" + "nature-inspired computing" -"deep learning"
- **Query 3:** "transfer learning" + "bio-inspired computing" -"deep learning"
- **Query 4:** "transfer learning" + "bio-inspired algorithms" -"deep learning"
- **Query 5:** "transfer learning" + "evolutionary computing" -"deep learning"
- **Query 6:** "transfer learning" + "evolutionary algorithms" -"deep learning"
- **Query 7:** "transfer learning" + "swarm intelligence" -"deep learning"
- **Query 8:** "transfer learning" + "artificial immune systems" -"deep learning"

Deep learning papers were excluded to focus the review specifically on transfer learning applications in evolutionary and swarm-based algorithms, avoiding the broad and distinct literature on TL in deep learning, which is already extensively reviewed elsewhere (Barthi

Table 1 Primary outcomes of the search queries for the IEEE, scopus, and web of science databases

Query	Search Engine		
	IEEE	Scopus	Web of Science
1	1	57	17
2	1	18	13
3	7	10	21
4	1	42	19
5	63	115	243
5	218	1233	284
7	25	446	96
8	15	117	9

Table 2 PICOS strategy

Criteria	Definition
P: Population	Nature-inspired algorithms
I: Intervention	Transfer learning
C: Comparator	State-of-art NIAs
O: Outcome	Performance (higher agility, better efficiency, etc)
S: Study design	Journals, articles, and conference papers on natural computing field

et al. 2020; Young et al. 2018). This restriction improves the focus of the survey and prevents overlap with existing reviews.

Table 1 summarizes the primary results of the searches in academic databases, totaling 2,380 studies collected by the eight strings in three academic databases.

3.2 Research sources and software

The search and retrieval of papers were managed using StArt Fabbri et al. (2016), a free systematic review management software developed at the Software Engineering Research Laboratory of the Federal University of São Carlos (LAPES-UFSCar). StArt was selected due to its robust study management capabilities, ease of data entry, and detailed information extraction. The review targeted three major databases: Scopus, IEEE Xplore, and Web of Science, chosen for their comprehensive coverage and indexing of high-quality journals. The combination of these sources minimizes publication bias and ensures that the review encompasses both computer science and interdisciplinary research outlets.

The PICOS framework Amir-Behghadami and Janati (2020) guided the study selection process, providing a structured and objective basis for defining inclusion and exclusion criteria, as shown in Table 2.

3.3 Exclusion criteria

After retrieving the initial set of articles, exclusion criteria were applied to refine the dataset. Searches were limited to publications from 2018 to 2023 within Scopus, IEEE Xplore, and Web of Science. Articles were excluded if they met any of the following conditions:

- Not peer-reviewed.

- In prepublication status.
- Duplicate entries.
- Published before 2018.
- Written in languages other than English.
- Data collected without ethics committee approval, when applicable.
- Did not address transfer learning in at least one of the targeted nature-inspired algorithms.
- Focused on transfer learning in deep learning.

Figure 4 depicts the PRISMA flowchart illustrating the study selection process. The initial search yielded 2,380 studies. Based on title and abstract screening, 291 studies were retained for further analysis. Subsequently, 40 duplicates and 204 studies failing one or more exclusion criteria were removed. After full-text reading and analysis, 47 articles were included in the final review.

3.4 Justification of methodological efficiency

The systematic review methodology was designed to ensure transparency, reproducibility, and comprehensiveness. Following PRISMA guidelines minimizes bias and enhances the quality of the evidence synthesis. The adoption of the PICOS framework provides a clear and structured approach to defining search terms, ensuring high relevance and reducing noise in the retrieved studies.

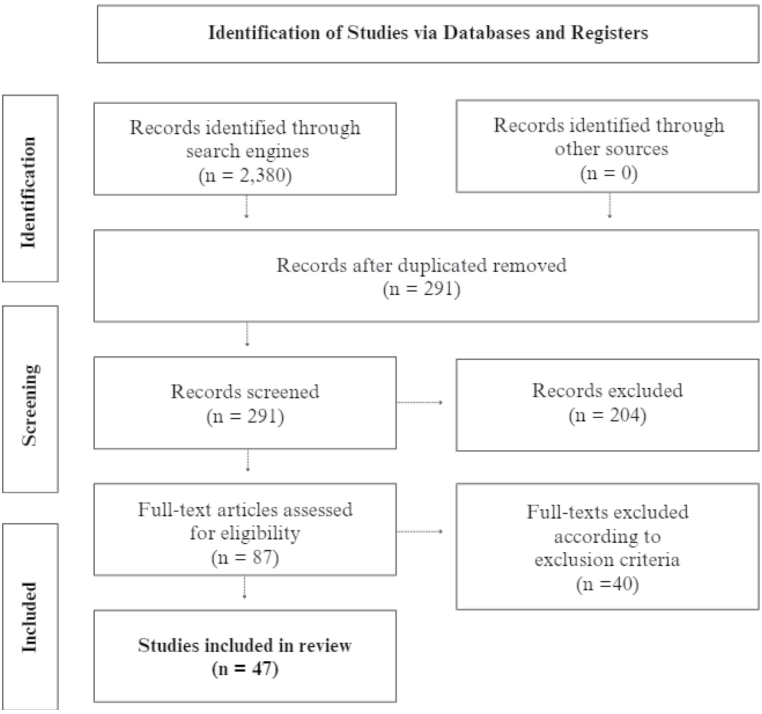


Fig. 4 PRISMA flow diagram

Utilizing the StArt tool improves methodological rigor by automating the management of inclusion/exclusion criteria, documenting reviewer decisions, and enabling consistent quality assessment. This reduces human error and enhances reliability throughout the review process.

The multi-stage filtering–comprising title screening, abstract review, and full-text evaluation–was performed by two independent reviewers to increase inter-rater reliability. Discrepancies were resolved through discussion, further strengthening the study's rigor. As a result, the final dataset not only reflects the state of the art but also provides a reliable foundation for analyzing transfer learning strategies in NIAs.

Together, these methodological components contribute to an efficient and high-quality synthesis of existing literature, offering a clear and trustworthy mapping of how transfer learning has been applied in nature-inspired algorithms.

4 Systematic review

This review aims to identify how transfer learning has been applied in nature-inspired algorithms, the goals behind its use in the case studies, and the main problems addressed by these algorithms. By systematically analyzing the studies, we also aim to highlight recurring patterns, challenges, and potential gaps in the integration of TL with NIAs, providing a critical overview rather than a mere listing of methods.

The search results show that transfer learning is predominantly employed in evolutionary computing and swarm intelligence algorithms. Among the evolutionary algorithms found are Multiobjective Evolutionary Algorithms (MOEA) and variants such as MOEA/D and SMOEA, Multifactorial Evolutionary Algorithm (MFEA), Multiobjective Evolutionary Clustering with Decomposition (MOEC/D), Genetic Programming (GP) and variants including GPHH and SGP, Genetic Algorithm (GA), Evolution Strategies (ES), and Differential Evolution (DE). Swarm intelligence algorithms identified include Particle Swarm Optimization (PSO), Artificial Bee Colony (ABC), and Ant Colony Optimization (ACO), as summarized in Table 3. These findings indicate a clear preference for well-established evolutionary and swarm-based algorithms when applying transfer learning, which may reflect both the maturity of these paradigms and their adaptability to knowledge transfer.

To answer the three research questions, this section is divided into three subsections:

- **Natural computing and transfer learning:** Discusses the research challenges of the transfer learning paradigm ("what to transfer?", "when to transfer?", and "how to transfer?") in nature-inspired algorithms. Emphasis is placed not only on the strategies employed but also on the reasoning behind the choice of components for transfer.
- **Objectives of transfer learning in natural computing algorithms:** Examines the aims behind applying transfer learning in the surveyed algorithms. Patterns and motivations, such as acceleration of convergence, improvement of solution quality, and reduction of computational costs, are highlighted.
- **Applications and performance evaluation:** Addresses the problems solved by nature-inspired algorithms combined with transfer learning, experimental setups, performance metrics, and comparative baselines. Insights into the practical impact of TL integration are provided, revealing which problem classes benefit most and under what conditions.

Table 3 Nature-inspired algorithms found in the base studies

Nature-inspired algorithms		
Evolutionary algorithms (EA)	Multiobjective evolutionary algorithms (MOEA)	Dynamic Multiobjective Evolutionary Algorithms (DMOEA) (Ye et al. 2022a; Ji et al. 2022; Yao and Wang 2023; Jiang et al. 2017; Zhao et al. 2023; Jiang et al. 2020a; Zhenzhong et al. 2019; Jiang et al. 2020b; Zhang et al. 2023a, b, c, d; He and Ren 2022; Aliniya and Khasteh 2023; Wu et al. 2023; Liu and Wang 2021; Jiang et al. 2020c; Zhenzhong et al. 2020)
		Multiobjective Evolutionary Algorithm Based on Decomposition (MOEA/D) Ye et al. (2022b), Fan et al. (2020)
		Static Multiobjective Optimization Evolutionary Algorithm (SMOEA) Wang et al. (2022a)
	Multifactorial evolutionary algorithm (MFEA) Lu et al. (2023), Ma et al. (2020)	
	Multiobjective evolutionary clustering with decomposition (MOEC/D) Liu et al. (2019)	
	Evolution strategy (ES) Lai et al. (2023), Chen and Liu (2021), Chen et al. (2021)	
	Differential evolution (DE) Huang et al. (2023), Chen et al. (2021)	
	Genetic programming (GP)	Genetic Programming Hyper Heuristics (GPHH) (Ardeh et al. 2021, 2020a, b, c, 2019; Russell and Pillay 2023; Chen et al. 2023)
		Standard Genetic Programming (SGP) (Fu et al. 2019; Iqbal et al. 2019; Chen et al. 2019; Muller et al. 2019; Hu et al. 2023)
	Genetic algorithm (GA) Zhang and Wang (2021)	
Swarm intelligence (SI)	Ant colony optimization (ACO) Singh and Pillay (2022)	
	Artificial bee colony (ABC) Zhou et al. (2023)	
	Particle swarm optimization (PSO) (Zheng et al. 2022; Lu et al. 2023; Huang et al. 2023; Fang et al. 2021; Huang and Chen 2021)	

4.1 Natural computing and transfer learning

The first question raised in this systematic review concerns which natural computing approaches use transfer learning. The focus here is on how transfer learning can be employed in the adaptation processes of nature-inspired algorithms to improve their performance. There are three main issues to consider before applying a transfer learning solution to an algorithm: "What to transfer?", "When to transfer?", and "How to transfer?", as previously discussed.

Regarding "what to transfer?", the concern lies in understanding which fragment of knowledge can be reused between tasks or domains Pan and Yang (2009). While some knowledge is specific to a particular domain or task, some components may be common to both, which can help improve performance. Identification of these transferable elements is critical to avoid ineffective or counterproductive knowledge transfer. Once transferable knowledge is identified, it is necessary to determine how it will be applied from one domain or task to another. Regarding "when to transfer?", it is important to investigate under which conditions knowledge transfer should occur. These considerations help prevent negative transfer and ensure that TL strategies are contextually appropriate.

It is equally important to identify situations in which knowledge transfer should not be performed. If these aspects are neglected, performance can be compromised, potentially resulting in negative transfer, where knowledge reuse degrades the target task's outcomes. Empirical evidence demonstrates that if two tasks are very dissimilar, indiscriminate transfer can harm the performance of the target tasks Rosenstein et al. (2005). This highlights the necessity of evaluating task and domain similarity prior to TL application in NIAs.

Two main groups of Nature-Inspired Algorithms (NIA) were detected as employing transfer learning solutions, totaling 49 articles (base studies). Of these, 42 papers focus on Evolutionary Computing, as shown in Table 4, and 7 are based on Swarm Intelligence, as shown in Table 5. Among the retrieved works, some papers were authored by the same research groups, with six different groups producing multiple works, in addition to groups with only one study on a given topic.

Most studies applied transfer learning techniques in controlled scenarios or simulations, often avoiding more complex or uncertain real-world applications. While several proposals focused on adapting transfer mechanisms, few presented robust comparative analyses or ablation studies to validate the effectiveness of the transferred knowledge. The lack of comprehensive validation in real-world contexts limits the generalization of reported results and indicates an important avenue for future research.

In the following subsections, evolutionary and swarm intelligence algorithms are addressed, analyzing each of the transfer learning issues. This systematic approach allows for a clear understanding of TL implementation and its impact, complementing the information presented in Tables 4 and 5.

4.1.1 What to transfer?

One crucial aspect in proposing the application of TL techniques is understanding "what to transfer" – that is, determining the knowledge to be transferred between source and target domains and tasks. During the analysis of the selected studies, both commonalities and divergences in the proposed algorithms were identified.

When addressing research problems in TL, particularly the question "what to transfer?", MOEA algorithms focus on transferring features and parameters (Ye et al. 2022a; Ji et al. 2022; Yao and Wang 2023; Jiang et al. 2017; Zhao et al. 2023; Jiang et al. 2020a; Zhenzhong et al. 2019; Jiang et al. 2020b; Zhang et al. 2023a, b, c, d; He and Ren 2022; Aliniya and Khasteh 2023; Wu et al. 2023; Liu and Wang 2021; Jiang et al. 2020c; Zhenzhong et al. 2020; Ye et al. 2022b; Fan et al. 2020; Wang et al. 2022a; Lu et al. 2023; Ma et al. 2020; Liu et al. 2019). In contrast, GP algorithms adopt strategies such as transferring design and code fragments, including reweighting instances and parameters, according to the algorithm's

Table 4 Research issues of evolutionary algorithms in the base studies

Evolutionary algorithms (EA)	Proposed algorithm	Transfer learning issues		
		What to transfer?	When to transfer?	How to transfer?
Genetic algorithms (GA)	Transfer Component Analysis Non-Dominated Sorting GA III (TC-NSGAIII) Zhang and Wang (2021)	Promising solutions and decision space distributions	During the initialization phase	Combining centroid distance and traditional TL
Evolution strategy (ES)	TL-Based Parallel Lower-Level Search EA with Covariance Matrix Adaptation ES (TLEA- CMA-ES) Chen et al. (2022)	Useful features from the neighbors	During the parallel optimization process	Sharing the best features during the process
	Improved TL Covariance Matrix Adaptation ES (ITLEA- CMA-ES) Lai et al. (2023)	The most similar tasks for knowledge transfer	A dynamic adjustment helps determine the optimal moments for transferring knowledge	Combining the proposed strategies (task selection and adaptive tuning) with TL framework
Differential evolution (DE)	TL-Based Algorithm with Differential Evolution (TLEA-DE) Chen et al. (2022)	Useful features from the neighbors	During the parallel optimization process	Sharing the best features during the process
	TL with Differential Evolution (TLDE) Huang et al. (2023)	Two learning according to the environment change	When the environment changes	Uses the adaptive parameter setting strategy
Multi objective evolutionary clustering with decomposition (MOEC/D)	TL-Assisted Multiobjective Evolutionary Clustering Framework w/ Decomposition (TrMOEC/D) Liu et al. (2019)	Learned patterns, cluster assignments and other relevant features	During the optimization process	Using an evolutionary clustering approach
Multifactorial EA (MFEA)	Model Transfer Multifactorial EA (MT-MFEA) Lu et al. (2023)	Genetic information among the DT and TT	During the evolutionary process as the algorithm adapts to changing problem conditions	Integrating clustering, traditional TL and local optima escaping strategy
	Two-Level TL (TLTLA) Ma et al. (2020)	Similar component tasks	During the evolutionary process as individuals evolve	Combining inter-task transfer and intra-task transfer
Dynamic multi objective EA (DMOEA)	Memory- Driven Manifold TL DMOEA (MMTL-DMOEA) Jiang et al. (2020a)	Information related to optimal individuals	During the evolutionary process	With Memory Mechanism and Manifold TL techniques
	TL-Based Surrogate Assisted DMOEA (TrSA- DMOEA) Zhang et al. (2023a)	A small number of previous high-quality knee solutions	During the evolutionary process	It uses statistical ML approaches to construct a prediction model
	Adaptive MOEA with Bilayer Selection (EADaBS) Zhao et al. (2023)	Information about promising solutions	During the evolutionary process	Combining solutions from different populations with an Ensemble

Table 4 (continued)

Evolutionary algorithms (EA)	Proposed algorithm	Transfer learning issues		
		What to transfer?	When to transfer?	How to transfer?
	DMOEA Based on Environmental Selection and TL (DMOEAESTL) He and Ren (2022)	Information from past optimization runs	During the initialization phase of the evolutionary algorithm	Integrating the historical data into the algorithmic framework
	Combinational Response Mechanisms DMOEA (CRM- DMOEA) Aliniya and Khasteh (2023)	Instead of transferring entire populations, it focuses on individual solutions	During the initialization phase of the optimization process	The individual based TL technique is applied
	DMOEA Based on TL and Elitism-Based Mutation (ATM- DMOEA) Zhang et al. (2023b)	Knowledge related to problem solving strategies	During the optimization process	Combining manifold TL mutation-based adaptation
	Cluster-Based Regression TL-Based DMOEA (CRTL-DMOEA) Zhang et al. (2023c)	Clusters centroids with POS	TLis triggered when environmental changes occur	Using cluster-based regression
	Elitism-Based TL and Diversity Maintenance DMOEA (HETD- DMOEA) Zhang et al. (2023d)	High-quality individuals from the previous search experience	Before the optimization process begins	Applying individual- based TL to enhance convergence speed and solution quality
	Regression TL Prediction-Based DMOEA (RTLP-DMOA) Zhenzhong et al. (2019)	Knowledge from historical populations obtained during previous optimization runs	When an environmental change is detected	Constructing a regression prediction model based on historical population data
	Dynamic Multiobjective EA (TPS- DMOEA) Liu and Wang (2021)	High-quality individuals	During the initialization phase of the optimization process	An individual based TL to accelerate construction of the initial population
	Individual Transfer-Based DMOEA (IT- DMOEA) Jiang et al. (2020b)	High-quality individuals	During the initialization phase of the optimization process	An individual based TL to accelerate construction of the initial population
	Knee Point-Based TL Method DMOEA (KT- DMOEA) Jiang et al. (2020c)	Knee points as the individuals that exhibit a trade-off between conflicting objectives	Dynamically during the optimization process	Using trend prediction models and imbalance TL
	Knee Points-Based Transfer DMOEA (KT- MCDM) Zhenzhong et al. (2020)	Knee points of the individuals and population	When predicting knee solutions for the next optimization step	Combining knee solutions and traditional TL
	TL Method Based On Clustering Difference DMOEA (TCD- DMOEA) Yao and Wang (2023)	Relevant knowledge about the population quality and solution space	During the initialization phase of the optimization process	Uses an approach based on clustering difference
	Multiple Source TL DMOEA (MSTL- DMOEA) Ye et al. (2022a)	Knowledge of the POS from the previous environment	At the beginning of each generation	Multiple source TL method with two procedures

Table 4 (continued)

Evolutionary algorithms (EA)	Proposed algorithm	Transfer learning issues		
		What to transfer?	When to transfer?	How to transfer?
	TL-Based DMOEA (T-DMOEA) Wu et al. (2023)	Knowledge related to high-quality solutions	Whenever an environmental change is detected	Manifold TL technique
	TL-Based DMOEA (Tr-DMOEA) Jiang et al. (2018)	Historical data of the previous DMOP	Before the evolutionary optimization process and during the adapting process	By using a domain adaptation technique
Multi objective EA (MMOEA)	Zoning Search and TL-Based MMOEA (ZSTL- MMOEA) Ji et al. (2022)	Feature representation from the most similar subspace to the current subspace	When the current subspace is similar enough to the most similar subspace	By using a method that is based on a cluster difference DMOP
Static multi objective EA (SMOEA)	Key Points-Based TL with Hybrid Prediction Strategies (KPTHP) Wang et al. (2022a)	Information related to the regional centroids of approximate POS	During the transition from one environment to another	With a clustering-based auto encoding approach
Multi objective EA based on decomposition (MOEA/D)	TL-Based MOEA Based on Decomposition w/ Efficient Global Optimization (TL-MOEA/ D-EGO) Fan et al. (2020)	Knowledge from the previous environment to the current one	After each change in the dynamic environment	Maps the previous training data to the current landscape using a linear transformation
	Hierarchical Clustering and Autoencoder Based DMOEA (CAE-DMOEA) Ye et al. (2022b)	Information related to the population quality and data distribution	During the initialization phase of the optimization process	Combining clustering and traditional TL
Standard genetic programming (SGP)	GP with Self-Learning Linear Model (sGP+SLLM) Fu et al. (2019)	Best GP program from the SD to the TD as the initial classifier	Before the training of the new classifier	By using a GP to optimize the weights of the classifier
	TL Genetic Programming Criptor (TLGP- Criptor) Iqbal et al. (2019)	The code fragments that are evolved by GP on a source image dataset	Before the evolutionary process of GP on the TD	Selects the best ones from the source population based on their fitness and diversity
	GP with TL (GPTL) Chen et al. (2019)	A new instance weighting for the TD	When the SD has sufficient weighted training data	By reweighting to put it on the TD
	TL-Based Surrogate GP (TL-SGP) Hu et al. (2023)	Hyper- Parameters, covariance structures and pre-trained model parameters	During the training phase	Using balance exploration (sampling uncertain regions) and exploitation (sampling promising regions)

Table 4 (continued)

Evolutionary algorithms (EA)	Proposed algorithm	Transfer learning issues		
		What to transfer?	When to transfer?	How to transfer?
	Primitives from Sub-Trees (PST) Muller et al. (2019)	Potentially good and unique building blocks from a source problem	During the initialization or preliminary stages of the GP process	With a building block selection mechanism
Genetic programming hyper heuristic (GPHH)	GP-Based TL Methods Ardeh et al. (2019)	Parameters of the routing policy learned by the GPHH	After the GPHH has converged on the SD and before the fine-tuning process on the TD	By using a local search operator that modifies the terminals of policy tree
	GP Hyper Heuristic with Probabilistic Prototype Tree Transfer (GPHH-PPT) Ardeh et al. (2020a)	Previous UCARP scenarios represented by a PPT	When initializing a new population for the TD	Using the PPT as a feature representation for the TD
	GP Hyper Heuristic with Surrogate Assisted Unique Full Tree Transfer (GPHH SUFullTree) Ardeh et al. (2021)	The phenotypes of the GP individuals in the SD	At the initialization stage of the GP algorithm in the TD	Applying a surrogate- assisted method
	Probabilistic Knowledge Enabled GP Hyper Heuristics (PKGPHH) Ardeh et al. (2020b)	The top members of a best population	In the initialization phase of the algorithm	With the Probabilistic Prototype Tree (PPT) technique
	Selection Perturbative Hyper Heuristic for TL (SHeTL) Russell and Pillay (2023)	Information such as promising solutions or problem specific insights	Dynamically decides when to transfer knowledge	With selection hyper-heuristic approach
	Diversity Oriented Knowledge Transfer for GP Hyper Heuristics (DDGPHH) Ardeh et al. (2020c)	Diverse and useful knowledge from the source domain	During the later training stages	Combining hypermutation with GPHH
	GP Hyper Heuristics for Multiobjective Location-Aware Service Brokering Problem (GPHH- MOLSB) Chen et al. (2023)	The group of POS heuristics from one domain to other	During the process of generating heuristics	Using TL into the GPHH framework to adapt heuristics

structure and objectives (Ardeh et al. 2021, 2020a, b, c, 2019; Russell and Pillay 2023; Chen et al. 2023; Fu et al. 2019; Iqbal et al. 2019; Chen et al. 2019; Muller et al. 2019; Hu et al. 2023). For GA, the focus is on transferring parameters such as mutation and crossover rates between tasks Zhang and Wang (2021). ES and DE aim to transfer useful features among neighboring optimizers, from solutions to fitness values (Lai et al. 2023; Chen and Liu 2021; Chen et al. 2021; Huang et al. 2023; Chen et al. 2021).

Meanwhile, in SI algorithms such as PSO, ACO, and ABC, the transfer involves population information, like the knowledge embedded in the best pheromone maps left by ants (Zheng et al. 2022; Lu et al. 2023; Huang et al. 2023; Fang et al. 2021; Huang and Chen 2021; Singh and Pillay 2022; Zhou et al. 2023). This range of strategies demonstrates the

Table 5 Research issues of the swarm intelligence algorithms in the base studies

Swarm intelligence algorithms (SI)	Proposed algorithm	Transfer learning issues		
		What to transfer?	When to transfer?	How to transfer?
Ant colony optimization (ACO)	TL-Based ACO Generation Constructive Hyper Heuristic (TL-HACO) Singh and Pillay (2022)	Pheromone map	At the beginning of the optimization process	Adapt and utilize the pheromone map to guide the search behavior of the ants
Artificial bee colony (ABC)	TL-Based ABC Algorithm (TLABC) Zhou et al. (2023)	Efficient resource utilization	At the optimization phase	Using feature extraction
Particle swarm optimization (PSO)	TL-Based PSO (TL-PSO) Zheng et al. (2022)	Parameters of the historical TSP	Before the training of the new TSP	Updating the parameters of the historical TSP
	TLEA Framework Based on PSO (TLPSO) Huang et al. (2023)	Best initial population	When the environment changes	Uses the adaptive feature setting strategy
	TL-Based PSO (TLPSO) Huang and Chen (2021)	Best initial population	When the environment changes	Uses the adaptive feature setting strategy
	TL-Based Adaptative PSO (TAPSO) Yang and Li (2023)	Knowledge from the historical problems that have been solved by PSO	Before the algorithm starts the searching process	By adjusting the features of PSO
	Sequence Transfer-Based PSO Algorithm (ST-PSO) Fang et al. (2021)	Discriminative information	During the learning process and modeling adaptation	All transfer objectives are combined in a single fitness function

flexibility and adaptability of TL techniques across different algorithmic contexts and problem domains.

4.1.2 When to transfer?

Regarding "when to transfer?", it is crucial to understand the appropriate timing for the transfer. In DMOEA and SMOEA, strategies include knowledge transfer during pre-search, upon detection of environmental changes, or during the evolutionary process itself (Ye et al. 2022a; Ji et al. 2022; Yao and Wang 2023; Jiang et al. 2017; Zhao et al. 2023; Jiang et al. 2020a; Zhenzhong et al. 2019; Jiang et al. 2020b; Zhang et al. 2023a, b, c, d; He and Ren 2022; Aliniya and Khasteh 2023; Wu et al. 2023; Liu and Wang 2021; Jiang et al. 2020c; Zhenzhong et al. 2020; Wang et al. 2022a). Conversely, MOEA/D, MFEA, and MOEC/D approaches transfer knowledge after each detected change in the dynamic environment, using change detection mechanisms and window size parameters to control the frequency of transfer (Ye et al. 2022b; Lu et al. 2023; Ma et al. 2020; Fan et al. 2020; Liu et al. 2019).

GP algorithms perform transfers during the design phase, initialization, or in situations such as label scarcity, adapting the strategy to the algorithm's specific context (Fu et al. 2019; Iqbal et al. 2019; Chen et al. 2019; Muller et al. 2019; Hu et al. 2023). In GPHH, transfers occur at convergence, before fine-tuning, at the start of each generation, or during initialization, depending on the algorithm's objective Ardeh et al. (2021, 2020a, 2020b, 2020c, 2019); Russell and Pillay (2023); Chen et al. (2023). GA, DE, and ES typically transfer knowledge during parallel optimization processes or prior to starting optimization on target problems (Zhang and Wang 2021; Huang et al. 2023; Chen et al. 2021; Lai et al. 2023; Chen and Liu 2021; Chen et al. 2021).

Overall, the timing of knowledge transfer is determined by both the algorithm's design and the characteristics of the problem domain, aiming to minimize negative transfer and maximize efficiency.

4.1.3 How to transfer?

As for "how to transfer?", it is necessary to understand the techniques used to perform TL. Generally, each algorithm in the base studies follows a specific strategy, reflecting the diversity of techniques within transfer learning and the variety of problems they aim to solve. In some MOEAs, transfer occurs sequentially: from the previous (source) environment to the current one, and then from the current to the next (target) environment (Ye et al. 2022a; Ji et al. 2022; Yao and Wang 2023; Jiang et al. 2017; Zhao et al. 2023; Jiang et al. 2020a; Zhenzhong et al. 2019; Jiang et al. 2020b; Zhang et al. 2023a, b, c, d; He and Ren 2022; Aliniya and Khasteh 2023; Wu et al. 2023; Liu and Wang 2021; Jiang et al. 2020c; Zhenzhong et al. 2020). This approach leverages the adaptive nature of the algorithms. Other MOEAs aim to minimize the negative effects of transfer by improving feature representation, similarly to MFEA (Ye et al. 2022b; Fan et al. 2020; Wang et al. 2022a; Lu et al. 2023; Ma et al. 2020; Liu et al. 2019).

GPHH algorithms use genetic operators like crossover and mutation on selected routing policies, replacing the worst policies with the best newly generated ones (Ardeh et al. 2021, 2020a, b, c, 2019; Russell and Pillay 2023; Chen et al. 2023). DE and ES adopt more dynamic strategies, sharing the best features throughout the evolutionary process (Huang

et al. 2023; Chen et al. 2021; Lai et al. 2023; Chen and Liu 2021; Chen et al. 2021). PSO algorithms demonstrate adaptive dynamic parameter adjustments in variable environments, similar to ACO and ABC algorithms (Zheng et al. 2022; Lu et al. 2023; Huang et al. 2023; Fang et al. 2021; Huang and Chen 2021; Zhou et al. 2023; Singh and Pillay 2022).

This diversity of TL strategies–including sequential transfer, feature representation improvement, dynamic sharing, and parameter adaptation–reflects the complexity required to optimize knowledge transfer across different algorithms and problem contexts.

4.2 Goals of TL in nature-inspired algorithms

Recent studies have demonstrated a growing interest in applying NIAs to functional and real-world optimization problems. Enhanced PSO variants have been proposed for global optimization Lu et al. (2023), while hybrid approaches such as ACO combined with CNNs have proven effective in medical diagnosis tasks like autism spectrum classification Ashraf et al. (2025). Additionally, algorithm refinements such as the Fire Hawks Optimizer Ashraf et al. (2023) and bat algorithm variants Bangyal et al. (2018) have advanced convergence and solution diversity. Research on mutation strategies in PSO has also contributed to practical applicability Bangyal et al. (2020). These developments emphasize the relevance of improving NIA adaptability and efficiency through TL.

When integrated with NIAs, TL enables benefits that surpass conventional optimization, such as reusing knowledge across tasks, adapting to dynamic problems, and improving solutions in low-data scenarios. These advantages are particularly relevant in real-world applications where data availability and computational resources are limited (Ardeh et al. 2020a, b, c, 2019, 2021).

Among the most significant benefits are the ability to reuse knowledge from past tasks to reduce search time; improved adaptability to dynamic and multi-scenario problems; greater efficiency in solving tasks with limited or costly data; and a reduction in the algorithm's dependency on random initialization. These advantages are especially important in real-world scenarios where resource constraints, data scarcity, or dynamic environments demand fast and adaptive solutions.

Understanding how TL is being incorporated into NIAs helps reveal which strategies are more promising and where further innovation is possible (Weiss et al. 2016; Zhuang et al. 2020; Pan and Yang 2009).

The second question raised in this review concerns understanding the problems addressed within nature-inspired algorithms. In the context of natural computing, the objective of using transfer learning varies depending on the algorithm and the application. This objective is directly related to the transfer and learning strategies described in Sects. 4.1.1 and 4.1.2. In the following sections, we describe which of these strategies were utilized in the algorithms proposed in the base studies.

4.2.1 Transfer strategies

The analysis of the base studies reveals that most proposed algorithms rely on three main strategies: Instance-Transfer, Feature-Representation-Transfer, and Parameter-Transfer, as summarized in Table 6.

Table 6 Transfer strategies and proposed algorithms

Transfer strategy	Proposed algorithm
Instance-transfer	MT-MFEA Lu et al. (2023), TrSA-DMOEA Zhang et al. (2023a), DMOEAESTL He and Ren (2022), CRM-DMOEA Aliniya and Khasteh (2023), TL-HACO Singh and Pillay (2022), CRTL-DMOEA Zhang et al. (2023c), HETD-DMOEA Zhang et al. (2023d), RTLP-DMOA Zhenzhong et al. (2019), TPS-DMOEA Liu and Wang (2021), IT-DMOEA Jiang et al. (2020b), GPTL Chen et al. (2019), KT-DMOEA Jiang et al. (2020c), KT-MCDM Zhenzhong et al. (2020), MSTL-DMOEA Ye et al. (2022a), PST Muller et al. (2019)
Feature-representation-transfer	MMTL-DMOEA Jiang et al. (2020a), EADaBS Zhao et al. (2023), ITLEA-CMA-ES Lai et al. (2023), PKGPHH Ardeh et al. (2020b), SHeTL Russell and Pillay (2023), ATM-DMOEA Zhang et al. (2023b), TAPSO Yang and Li (2023), TLTLA Ma et al. (2020), CAE-DMOEA Ye et al. (2022b), KPTHP Wang et al. (2022a), DDGPHH Ardeh et al. (2020c), sGP+SLLM Fu et al. (2019), GPHH-PPT Ardeh et al. (2020a), TLGP-Criptor Iqbal et al. (2019), TC-NSGAIII Zhang and Wang (2021), GPHH-MOLSB Chen et al. (2023), ST-PSO Fang et al. (2021), TLABC Zhou et al. (2023), TL-MOEA/D-EGO Fan et al. (2020), GPHH SUFullTree Ardeh et al. (2021), Tr-DMOEA Jiang et al. (2018), TL-CMA-ES Chen and Liu (2021), TLDE Huang et al. (2023), TLPSO Huang et al. (2023), TLPSO Huang and Chen (2021), TCD-DMOEA Yao and Wang (2023), TLEA-CMA-ES Chen et al. (2022), TLEA-CMA-DE Chen et al. (2022), GP-Based TL Methods Ardeh et al. (2019), ZSTL-MMOEA Ji et al. (2022)
Parameter-transfer	TL-PSO Zheng et al. (2022), T-DMOEA Wu et al. (2023), TL-SGP Hu et al. (2023), TLABC Zhou et al. (2023), TrMOEC/D Liu et al. (2019)

The *Instance-Transfer* strategy appears in 16 algorithms, spanning Evolutionary Computing (MFEA, DMOEA, GP, GPHH) and Swarm Intelligence (ACO). In the DMOEA category, key examples include:

- TrSA-DMOEA Zhang et al. (2023a), which builds predictive models based on previously solved instances.
- DMOEAESTL He and Ren (2022), which reuses POS solutions to preserve diversity and accelerate convergence.
- CRM-DMOEA Aliniya and Khasteh (2023), which prevents negative transfer by preserving valuable knowledge.
- CRTL-DMOEA Zhang et al. (2023c), which adapts after environmental changes.
- HETD-DMOEA Zhang et al. (2023d), which transfers high-quality individuals.
- RTLP-DMOA Zhenzhong et al. (2019), which enhances initial populations using historical instances.

Within Evolutionary Computing, three Standard Genetic Programming representatives stand out: PST Muller et al. (2019), transferring building blocks between problems; GPTL Chen et al. (2019), focusing on knowledge transfer in optimization tasks; and TL-SGP Hu

et al. (2023), following a similar trend. The MFEA-based algorithm, MT-MFEA Lu et al. (2023), transfers elite individuals from past instances to initialize new populations, guiding search processes.

In Swarm Intelligence, TL-HACO Singh and Pillay (2022) (ACO-based) demonstrates instance reuse across domains within the same resource space. These examples collectively illustrate diverse approaches to instance-level knowledge transfer, highlighting both efficiency gains and potential challenges related to task similarity and negative transfer.

The *Feature-Representation-Transfer* strategy is adopted in 30 algorithms, covering a wide range of techniques. In MFEA, TLTLA Ma et al. (2020) emphasizes feature-level knowledge reuse. In SMOEA, KPTHP Wang et al. (2022a) encodes POS centroids into transferable features. Within DMOEA, several algorithms stand out:

- MMTL-DMOEA Jiang et al. (2020a) and EADaBS Zhao et al. (2023) use shared representations for multi-objective optimization.
- ATM-DMOEA Zhang et al. (2023b) and CAE-DMOEA Ye et al. (2022b) transfer population quality and data distribution.
- Tr-DMOEA Jiang et al. (2018) handles scenario dynamics.
- TCD-DMOEA Yao and Wang (2023) and ZSTL-MMOEA Ji et al. (2022) focus on improving initial populations through feature reuse.

In GP, algorithms such as sGP+SLLM Fu et al. (2019), TLGP-Criptor Iqbal et al. (2019), and TC-NSGAIII Zhang and Wang (2021) utilize feature representations for model building and optimization. GPHH-based methods including PKGPHH Ardeh et al. (2020b), SHeTL Russell and Pillay (2023), DDGPHH Ardeh et al. (2020c), GPHH-PPT Ardeh et al. (2020a), GPHH SUFullTree Ardeh et al. (2021), GPHH-MOLSB Chen et al. (2023), and GP-Based TL Methods Ardeh et al. (2019) leverage various strategies to improve transfer efficiency and adaptability across tasks.

In the ES category, TL-CMA-ES Chen and Liu (2021), TLEA-CMA-ES Chen et al. (2022), and ITLEA-CMA-ES Lai et al. (2023) transfer knowledge via adapted covariance matrices. DE-based methods such as TLDE Huang et al. (2023) and TLEA-DE Chen et al. (2021) follow a similar approach.

Within Swarm Intelligence-based algorithms, Particle Swarm Optimization algorithms include ST-PSO Fang et al. (2021), transferring common features between domains, TLPSO Huang and Chen (2021), facilitating knowledge transfer, and TAPSO Yang and Li (2023), dynamically adapting for efficient knowledge transfer in variable environments.

The *Parameter-Transfer* strategy is applied in five algorithms, emphasizing parameter reuse across tasks. In MOEA, T-DMOEA Wu et al. (2023) dynamically adapts its parameters for evolving scenarios. TrMOEC/D Liu et al. (2019) integrates TL to improve parameter adaptation and automation. In Swarm Intelligence, TL-PSO Zheng et al. (2022) leverages past parameters for optimization. In ABC, TLABC Zhou et al. (2023) combines both parameter and feature transfer, aiming for cross-domain generalization.

Overall, these strategies demonstrate that parameter-level adaptation can enhance TL performance in Nature-Inspired Algorithms by reducing retraining costs, increasing adaptability, and mitigating challenges associated with domain/task similarity and negative transfer.

4.2.2 Learning strategies

Transfer learning comprises three strategies: *Inductive-Transfer, Transductive-Transfer*, and *Unsupervised-Transfer*. The analysis of the base studies indicates that most proposed algorithms follow the inductive strategy, followed by four transductive and eleven unsupervised algorithms, as summarized in Table 7.

The *Inductive Learning* strategy is the most prevalent, represented by 36 algorithms. Among MOEA algorithms, MMTL-DMOEA Jiang et al. (2020a), KT-DMOEA Jiang et al. (2020c), and Tr-DMOEA Jiang et al. (2018) leverage historical data for effective transfer. T-DMOEA Wu et al. (2023) adapts parameters based on past tasks to maintain performance in dynamic environments. TrSA-DMOEA Zhang et al. (2023a), ATM-DMOEA Zhang et al. (2023b), CRTL-DMOEA Zhang et al. (2023c), and HETD-DMOEA Zhang et al. (2023d) use past experiences to guide current optimization. EADaBS Zhao et al. (2023) and RTLP-DMOA Zhenzhong et al. (2019) construct predictive models from prior popula-

Table 7 Learning strategies and proposed algorithms

Learning strategy	Proposed algorithms
Inductive	MMTL-DMOEA Jiang et al. (2020a), MT-MFEA Lu et al. (2023), TrSA-DMOEA Zhang et al. (2023a), EADaBS Zhao et al. (2023), ITLEA-CMA-ES Lai et al. (2023), SHeTL Russell and Pillay (2023), ATM-DMOEA Zhang et al. (2023b), TL-HACO Singh and Pillay (2022), TAPSO Yang and Li (2023), TL-PSO Zheng et al. (2022), CRTL-DMOEA Zhang et al. (2023c), DDGPHH Ardeh et al. (2020c), T-DMOEA Wu et al. (2023), HETD-DMOEA Zhang et al. (2023d), RTPL-DMOA Wang et al. (2020), sGP-SLLM Fu et al. (2019), GPHH-PPT Ardeh et al. (2020a), TLGP-Criptor Iqbal et al. (2019), GPTL Chen et al. (2019), KT-DMOEA Jiang et al. (2020c), GPHH-MOLSB Chen et al. (2023), MSTL-DMOEA Ye et al. (2022a), TL-SGP Hu et al. (2023), TLABC Zhou et al. (2023), TL-MOEA/D-EGO Fan et al. (2020), GPHH SUFullTree Ardeh et al. (2021), PST Muller et al. (2019), Tr-DMOEA Jiang et al. (2018), TLDE Huang et al. (2023), TLPSO Huang et al. (2023), TLPSO Huang and Chen (2021), TLEA-CMA-ES Chen et al. (2021), TLEA-CMA-DE Chen et al. (2021), GP-Based TL Methods Ardeh et al. (2019), ZSTL-MMOEA Ji et al. (2022)
Transductive	PKGPHH Ardeh et al. (2020b), ATM-DMOEA Zhang et al. (2023b), TPS-DMOEA Liu et al. (2019), TL-CMA-ES Chen and Liu (2021)
Unsupervised	DMOEAESTL He and Ren (2022), CRM-DMOEA Aliniya and Khasteh (2023), TLTLA Ma et al. (2020), CAE-DMOEA Ye et al. (2022b), KPTHP Wang et al. (2022a), TC-NSGAIII Zhang and Wang (2021), IT-DMOEA Liu et al. (2019), KT-MCDM Wang et al. (2020), ST-PSO Fang et al. (2021), TrMOEC/D Liu et al. (2019), TCD-DMOEA Yao and Wang (2023)

tions. MSTL-DMOEA Ye et al. (2022a) integrates knowledge from multiple sources, while ZSTL-MMOEA Ji et al. (2022) improves diversity and convergence in multimodal scenarios. TPS-DMOEA Liu and Wang (2021) generalizes past solutions for ongoing optimization. In MOEA/D, TL-MOEA/D-EGO Fan et al. (2020) reuses information from similar tasks to guide search. MT-MFEA Lu et al. (2023), in the MFEA category, transfers genetic information from source domains to improve target task performance. These examples highlight how inductive transfer can accelerate convergence and reduce the reliance on random initialization.

The *Transductive Learning* strategy aims to enhance performance in specific tasks by adapting knowledge from related ones. ATM-DMOEA Zhang et al. (2023b) dynamically adjusts optimization in response to environmental changes. TL-CMA-ES Chen and Liu (2021), based on ES, updates the population distribution via covariance matrix adaptation. PKGPHH Ardeh et al. (2020b), a GPHH-based method, captures complex patterns to enable effective transfer. TPS-DMOEA Liu and Wang (2021) also reflects this strategy. Collectively, these approaches emphasize efficiency gains, adaptability to dynamic problems, and the mitigation of negative transfer effects when tasks are similar.

The *Unsupervised Learning* strategy enables knowledge transfer without labeled data. In DMOEA algorithms, DMOEAESTL He and Ren (2022) and CRM-DMOEA Aliniya and Khasteh (2023) use historical data or individuals to improve performance. IT-DMOEA Jiang et al. (2020b) and TCD-DMOEA Yao and Wang (2023) adopt similar approaches. In SMOEA, KPTHP Wang et al. (2022a) maps relationships between prior POS centroids. CAE-DMOEA Ye et al. (2022b) and KT-MCDM Zhenzhong et al. (2020) identify patterns in the data distribution without labels. TrMOEC/D Liu et al. (2019), from the MOEC/D category, focuses on discovering data structures. In MFEA, TLTLA Ma et al. (2020) adapts evolutionary dynamics in multi-objective problems. In GAs, TC-NSGAIII Zhang and Wang (2021) applies transferred knowledge from earlier tasks to enhance search. ST-PSO Fang et al. (2021), in the SI context, guides swarm behavior using past experiences without requiring supervision. Unsupervised transfer provides flexibility in low-data scenarios but may face limitations in controlling negative transfer, highlighting areas for further research.

4.3 Applications targeted by the algorithms

The third research question pertains to the real-world problems and applications addressed by TL-enhanced NIAs. Table 8 summarizes the types of optimization problems, algorithm names, benchmarks, instances used, and performance metrics. This overview highlights the versatility and practical relevance of TL-based enhancements in nature-inspired algorithms.

Applications span multi-objective problems (MOPs), dynamic MOPs (DMOPs), many-objective problems (MMOPs), multi-scenario MOPs (MSMOPs), evolutionary optimization problems (EOPs), multi-fidelity optimization problems (MFOPs), discrete optimization (e.g., TSP, QAP, UCARP), bi-level optimization (BLOPs), function approximation, classification, and regression, demonstrating the broad applicability of TL-enhanced NIAs across diverse domains.

Each algorithm is benchmarked using established metrics (e.g., HV, IGD, MIGD, STD) and evaluated on specific problem instances. For example, TL-PSO and TL-CMA-ES are effective in discrete optimization, while TLGP-Criptor and SheTL focus on regression and classification, evaluated using accuracy and Wilcoxon tests. This variety reflects the ability

Table 8 Experiments information and proposed algorithms

Optimization type	Problem	Proposed algorithms	Performance measures
DMOPs	Standard DMOPs	MMTL-DMOEA Jiang et al. (2020a), DMOEAESTL He and Ren (2022), CRM-DMOEA Aliniya and Khasteh (2023), ATM-DMOEA Zhang et al. (2023b), CRTL-DMOEA Zhang et al. (2023c), CAE-DMOEA Ye et al. (2022a), KPTHPT-DMOEA Wang et al. (2022a), HETD-DMOEA Zhang et al. (2023d), TC-NSGAIII Zhang and Wang (2021), TPS-DMOEA Liu and Wang (2021), IT-DMOEA Jiang et al. (2020b), KT-DMOEA Jiang et al. (2020c), KT-MCDM Wang et al. (2020), MSTL-DMOEA Ye et al. (2022a), Tr-DMOEA Jiang et al. (2018), TCD-DMOEA Yao and Wang (2023), RTLP-DMOEA Wang et al. (2020)	IGD, MIGD, DMIGD, SP, MS, MHV, MMS, STD, KneeDist, Mean
	Expensive DMOPs	TrSA-DMOEA Zhang et al. (2023a)	MIGD, MMS, FR
MOPs	SMOPs	TL-MOEA/D-EGO Fan et al. (2020), TrMOEC/D Liu et al. (2019), TLDE Huang et al. (2023), TLPSO Huang and Chen (2021)	MIGD, RI, NMI
	MMOPs	ZSTL-MMOEA Ji et al. (2022)	PSP, HV
	MSMOPs	EADaBS Zhao et al. (2023)	IGD
	MTOs	TLTLA Ma et al. (2020)	Mean, STD
	EOPs	TL-SGP Hu et al. (2023)	Average Run Time, Average Number of Buildings, Convergence Curves, Normality, Sum of Squares, Mean Square, Variance Test, Accuracy
	MFOPa	MT-MFEA Lu et al. (2023)	Mean Function Values
Discrete optimization	UCARP	PKGPHH Ardeh et al. (2020b), DDGPHH Ardeh et al. (2020c), GPHH-PPT Ardeh et al. (2020a), GPHH SUFullTree Ardeh et al. (2021), SubTree Ardeh et al. (2019), FullTree Ardeh et al. (2019) BestGen Ardeh et al. (2019), TLGPCriptor Ardeh et al. (2019), GTLKnow Ardeh et al. (2019)	Number of "Wins", "Draws", "Losses", Convergence Curve, Friedman's Test, STF, Mean
	MSSP	TL-HACO Singh and Pillay (2022)	Mann–Whitney U
	1BBP	TL-HACO Singh and Pillay (2022)	Mann–Whitney U
	2DPP	ST-PSO Fang et al. (2021)	Best Time, Average Time
	QAP	TL-HACO Singh and Pillay (2022)	Mann–Whitney U
	GOPs	TAPSO Yang and Li (2023)	Convergence Analysis
	MOLSBs	GPHH-MOLSB Chen et al. (2023)	IGD, HV
	MCSAs	TLABC Zhou et al. (2023)	STD, Accuracy
	TSPs	TL-PSO Zheng et al. (2022)	BPL, APL, STD
Function approximation		TLPSO Huang and Chen (2021); Huang et al. (2023)	IGD
Classification		sGP+SLLM Fu et al. (2019), TLGP-Criptor Iqbal et al. (2019)	Mean, STD, Accuracy
Regression		SHeTL Russell and Pillay (2023), GPTL Chen et al. (2019), PST Muller et al. (2019)	BTF, MTF, Accuracy

of TL to enhance algorithm performance across both combinatorial and continuous optimization tasks.

Multi-objective problems (MOPs) involve optimizing conflicting objectives, requiring solutions that balance trade-offs instead of a single optimum. Foundational studies include MOEA/D-EGO Fan et al. (2020), TrMOEC/D Liu et al. (2019), TLDE Huang et al. (2023), and TLPSO Huang and Chen (2021), evaluated using metrics such as modified inverted generational distance (MIGD), Rand Index (RI), and normalized mutual information (NMI). For multi-scenario MOPs (MSMOPs), EADaBS Zhao et al. (2023) is the representative algorithm, with comparative experiments based on IGD.

Evolutionary optimization problems (EOPs), represented by TL-SGP Hu et al. (2023), are assessed using metrics such as average runtime, number of buildings, and convergence curves. For many-objective problems (MMOPs), ZSTL-MMOEA Ji et al. (2022) is evaluated using Pareto set proximity (PSP) and hypervolume (HV). Multi-task optimization problems (MTOPs), addressed by TLTLA Ma et al. (2020), are evaluated using mean and standard deviation (STD). Dynamic multi-objective problems (DMOPs), such as those tackled by TrSA-DMOEA Zhang et al. (2023a), use metrics like MIGD and modified maximum spread (MMS). These studies illustrate how TL supports adaptability in dynamic and multi-scenario environments.

Discrete optimization includes various domains. The one-dimensional bin packing problem (1BPP) is addressed by ITLEA-CMA-ES Lai et al. (2023), TL-CMA-ES Chen and Liu (2021), TLEA-CMA-ES Chen et al. (2022), and TLEA-CMA-DE Chen et al. (2023), with metrics such as median, interquartile range (IQR), and absolute deviation from the real optimum (ADRO). The two-dimensional packing problem (2DPP), handled by ST-PSO Fang et al. (2021), is evaluated using best and average solution times. TL-HACO Singh and Pillay (2022) addresses the quadratic assignment and movie scene scheduling problems, using metrics like p-value, z-score, and effect size. These evaluations indicate TL's role in enhancing convergence and maintaining solution diversity under complex scenarios.

For the traveling salesman problem (TSP), TL-PSO Zheng et al. (2022) evaluates These evaluations indicate TL's role in enhancing convergence and maintaining solution diversity under complex scenarios performance through best and average path length. The uncertain capacitated arc routing problem (UCARP) is addressed by various GP-based TL methods Ardeh et al. (2019), with monitored convergence curves and Friedman's test as evaluation tools. GPHH-MOLSB Chen et al. (2023) solves multi-objective large-scale bin packing (MOLSB), evaluated with IGD and HV. TLABC Zhou et al. (2023) targets multiple-choice subset assignment (MCSA), with evaluation based on STD and accuracy using Wilcoxon and Friedman tests. TAPSO Yang and Li (2023) tackles global optimization problems (GOPs) through convergence analysis.

Bi-level optimization problems are tackled by TL-CMA-ES Chen and Liu (2021), TLEA-CMA-ES Chen et al. (2022), ITLEA-CMA-ES Lai et al. (2023), and TLEA-DE Chen et al. (2022), using metrics such as ADRO, IQR, and median. In function approximation, TLPSO (Huang and Chen 2021; Huang et al. 2023) and TLGP-Criptor Iqbal et al. (2019) are evaluated via IGD. Classification tasks involve sGP+SLLM Fu et al. (2019) and TLGP-Criptor Iqbal et al. (2019), using accuracy, mean, and STD as performance indicators. For regression, SheTL Russell and Pillay (2023), GPTL Chen et al. (2019), and PST Muller et al. (2019) are evaluated using metrics such as best and median testing fitness, and the Wilcoxon signed-rank test.

Overall, these findings demonstrate that TL-enhanced NIAs provide measurable improvements in both optimization and learning-based tasks, though challenges such as domain similarity and negative transfer remain areas for further investigation.

4.4 Identified study gaps

Among the primary studies reviewed, it was possible to observe a trend in research towards evolutionary algorithms, with approximately 83% of the articles proposing this category of algorithms with TL implementation. In contrast, only 17% of the studies focus on swarm intelligence algorithms, indicating an opportunity for further investigation in this area.

Transfer learning in swarm intelligence algorithms remains an underexplored area. Strategies such as leveraging similar tasks or utilizing historical data have been rarely investigated. Moreover, various forms of transferable knowledge, such as pheromone maps and historical parameter settings, have been addressed in only a limited number of studies, highlighting the narrow scope of current research. Additionally, there is a notable absence of methods capable of dynamically determining the optimal timing for knowledge transfer, which limits adaptability in dynamic or real-world environments.

This study also reveals gaps in the application of TL in several optimization problems. There is a concentration of studies in DMOPs, while areas such as Discrete Optimization and Function Approximation are less explored. Furthermore, the limited variety of evaluation metrics in some domains impedes robust performance comparisons. In addition, the limited variety of metrics in some domains makes robust comparisons difficult. Specific problems, such as MSMOPs and EOPs, also have few studies, indicating research opportunities. The low application of TL in complex combinatorial problems, such as TSP, reinforces the need for further research to expand applicability and establish standardized evaluation protocols.

Analysis of transfer strategies indicates additional research opportunities. Parameter transfer is the least investigated approach, suggesting that future studies could explore its effectiveness and adaptability across different domains. Regarding learning strategies, unsupervised TL is also underrepresented, with only a few specific algorithms employing it, indicating significant potential for applying unsupervised transfer learning in complex and dynamic optimization scenarios.

Those gaps underscore the need for systematic research to diversify algorithm types, explore less-studied strategies, and expand the range of problem domains for TL-enhanced nature-inspired algorithms.

4.5 Meta and citation analysis

Quantitative analysis of the 47 selected studies shows a clear dominance of Evolutionary Algorithms (83%) over Swarm Intelligence (17%). Feature-Representation Transfer emerges as the most frequently employed strategy (61%), followed by Instance-Transfer (33%) and Parameter-Transfer (8%).

Feature-Representation Transfer is present in 61% of the primary studies, making it the most employed technique. Instance-Transfer appears in 33% of the studies, while Parameter-Transfer is used in only 8%. These results indicate that adapting and reusing feature representations is generally the most effective approach for knowledge transfer between tasks, particularly when input patterns differ substantially across domains.

Regarding learning strategies, Inductive Transfer predominates, appearing in 73% of the works, whereas Transductive Transfer and Unsupervised Transfer are much less frequent, at 6% and 2%, respectively. This pattern highlights a strong tendency to generalize knowledge from source tasks to related target problems, often without requiring explicit labeling or supervision.

Overall, the meta-analysis indicates a research landscape where evolutionary algorithms combined with Feature-Representation and Inductive Transfer strategies constitute the prevalent model for TL application in nature-inspired algorithms. This overview provides solid guidance for future studies, both for improving existing approaches and for exploring underrepresented areas, particularly swarm intelligence, Parameter-Transfer, and Unsupervised Transfer strategies.

Citation analysis based on Google Scholar data allowed identifying the most accessed proposed algorithms after publication, reflecting their relevance in the field.

Citation analysis based on Google Scholar data highlights the most referenced proposed algorithms after publication, reflecting their impact and relevance in the field. Within Swarm Intelligence algorithms, TL-PSO Zheng et al. (2022) is the most cited, with 38 citations since 2022, followed by ST-PSO with 17 citations. Among Artificial Bee Colony (ABC)-based algorithms, TLABC Zhou et al. (2023) stands out with 16 citations since 2023.

5 Conclusions and future trends

This paper presented a comprehensive systematic review on the application of Transfer Learning (TL) within Nature-Inspired Algorithms (NIAs), explicitly excluding deep learning approaches to focus on classical bioinspired optimization methods. The analysis covered 47 primary studies, examining how TL has been integrated into evolutionary and swarm-based algorithms across learning paradigms, transfer strategies, algorithmic goals, and problem domains.

The results demonstrate that TL can significantly enhance NIAs by improving generalization, accelerating convergence, reducing dependency on random initialization, and enabling knowledge reuse in data-scarce or dynamic environments. Key theoretical challenges remain, particularly concerning the core TL questions: what knowledge to transfer, when to transfer, and how to transfer it effectively. Mishandling these aspects may lead to negative transfer, performance degradation, or increased algorithmic complexity. Nevertheless, diverse strategies have emerged, highlighting the adaptability and interdisciplinary potential of TL in NIA contexts.

Practically, TL-enabled NIAs have shown advantages in solving complex problems such as multiobjective, dynamic, and discrete optimization, with growing applications in logistics, manufacturing, and healthcare. However, notable gaps persist, including limited research on clustering and combinatorial tasks, underrepresented unsupervised learning approaches, parameter transfer, and the lack of standardized evaluation protocols.

Future research should investigate novel TL mechanisms tailored for NIAs, especially in multitask and unsupervised settings. Emphasis is needed on effective feature transfer, domain adaptation, and hybrid model design. Additionally, integrating TL with underexplored paradigms, such as artificial immune systems and other bioinspired techniques, represents a promising avenue.

This review consolidates current efforts to combine TL with nature-inspired approaches, highlighting significant opportunities for advancement. By focusing on TL in NIAs without deep learning, this work lays a foundation for new hybrid algorithms capable of addressing real-world optimization challenges with enhanced adaptability and efficiency.

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Declarations

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